ChatGPT 5.2

infinite square well nth energy formula derive. full math, s
remember i know all this just forgot so refresher is neede

Fine. Here's your "I definitely knew this once" refresher.

1D infinite square well: setup

Potential:

$$V(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & \text{else} \end{cases}$$

So $\psi(0) = \psi(L) = 0$, and inside the well the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi.$$

Rearrange:

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0, \quad k^2 = \frac{2mE}{\hbar^2}.$$

General solution + boundary conditions

General solution:

$$\psi(x) = A\sin(kx) + B\cos(kx).$$

Apply $\psi(0) = 0$:

$$\psi(0) = A\sin 0 + B\cos 0 = B = 0 \Rightarrow B = 0.$$

So $\psi(x) = A\sin(kx)$.

Apply $\psi(L) = 0$:

$$\psi(L) = A\sin(kL) = 0.$$

Nontrivial $A \neq 0$ implies

$$\sin(kL) = 0 \Rightarrow kL = n\pi, \quad n = 1, 2, 3, \ldots$$

Thus

$$k_n = \frac{n\pi}{L}.$$

Energy quantization

Using $E = \frac{\hbar^2 k^2}{2m}$:

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2 = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots$$

That's the whole thing: boundary conditions force discrete k, which forces discrete E.

do for finite square well too and find tunnel prol

Sure. Humans really love putting particles in boxes and then acting shocked when the matl

I'll do **finite square well bound states** (the classic transcendental conditions), then a **tunn** **transmission probability** for a **finite barrier** (that's what people usually mean by "tunnel p The *well* itself has evanescent tails but no "transmission" unless you set up scattering.

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1) Finite square well (bound states)

Take the standard symmetric well:

$$V(x) = \begin{cases} 0, & |x| < a \\ V_0, & |x| \geq a \end{cases} \qquad (V_0 > 0)$$

Bound states satisfy $0 < E < V_0$.

Schrödinger solutions in each region

Inside ($|x| < a$, $V = 0$):

$$-\frac{\hbar^2}{2m}\psi'' = E\psi \;\Rightarrow\; \psi'' + k^2\psi = 0, \quad k = \frac{\sqrt{2mE}}{\hbar}.$$

So $\psi = A\cos(kx) + B\sin(kx)$.

Outside ($|x| \geq a$, $V = V_0$):

$$-\frac{\hbar^2}{2m}\psi'' + (V_0)\psi = E\psi \;\Rightarrow\; -\frac{\hbar^2}{2m}\psi'' = (E - V_0)\psi.$$

Since $E < V_0$, define

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar},$$

and the decaying solutions are

$$\psi(x) = Ce^{-\kappa|x|} \quad \text{(up to piecewise constants)}.$$

Use parity to simplify (symmetric well)

Even states: ψ even.

- Inside: $\psi(x) = A\cos(kx)$
- Outside for $x \geq a$: $\psi(x) = Fe^{-\kappa(x-a)}$

Match ψ and ψ' at $x = a$:

$$A\cos(ka) = F$$

$$-Ak\sin(ka) = -\kappa F$$

Divide the derivative condition by the value condition:

$$\frac{-Ak\sin(ka)}{A\cos(ka)} = \frac{-\kappa F}{F} \;\Rightarrow\; k\tan(ka) = \kappa.$$

Even-state quantization:

$$\boxed{k\tan(ka) = \kappa.}$$

Odd states: ψ odd.

- Inside: $\psi(x) = B\sin(kx)$
- Outside for $x \geq a$: $\psi(x) = Fe^{-\kappa(x-a)}$

Match at $x = a$:

$$B\sin(ka) = F$$

$$Bk\cos(ka) = -\kappa F$$

Divide:

$$\frac{Bk\cos(ka)}{B\sin(ka)} = \frac{-\kappa F}{F} \Rightarrow k\cot(ka) = -\kappa.$$

Equivalently:

$$\boxed{-k\cot(ka) = \kappa.}$$

Relationship to energy

Remember

$$k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}.$$

And once you solve the transcendental equation for k (or E), energy is

$$\boxed{E = \frac{\hbar^2 k^2}{2m}}, \quad 0 < E < V_0.$$

Useful dimensionless form (for quick solving):

$$z = ka, \quad p = \kappa a, \quad \rho^2 = z^2 + p^2 = \frac{2mV_0 a^2}{\hbar^2}.$$

Then

$$\text{even: } z\tan z = p, \qquad \text{odd: } -z\cot z = p.$$

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2) "Tunnel probability" (transmission through a finite barrier)

This is the standard tunneling setup:

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & 0 < x < L \\ 0, & x > L \end{cases}$$

Particle comes from the left with energy $E < V_0$. Define

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

The transmission probability T (exact, textbook result) is

$$\boxed{T = \frac{1}{1 + \frac{V_0^2}{4E(V_0-E)}\sinh^2(\kappa L)}} \quad (E < V_0).$$

Thick/high barrier limit ($\kappa L \gg 1$):

$$\sinh^2(\kappa L) \approx \frac{1}{4}e^{2\kappa L}$$

so

$$\boxed{T \approx \frac{16E(V_0 - E)}{V_0^2}\, e^{-2\kappa L}}$$

and the "physics in one glance" part is the exponential:

$$T \propto e^{-2\kappa L}, \quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}.$$

Quick mental map so you don't forget again

- **Finite well bound states:** oscillatory inside, exponential tails outside, **match at bound transcendental**.
- **Tunneling through barrier:** exponential inside barrier, but still finite **transmission** beca gives a nonzero outgoing wave → T above.

If you meant **penetration probability density** outside the finite *well* (tail weight) instead of transmission, that's a different quantity (integral of $|\psi|^2$ in $|x| > a$), and it comes directly f κ tail.

wait how can solution have a constant E? isn't E an unkn really forgot this stuff

Relax. Nothing is broken except the memory cache.

The confusion is **exactly** the right one to have, so let's clean it up fast and cleanly.

The key idea you forgot

In the **time-independent Schrödinger equation**,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi,$$

E **is an eigenvalue**, not an input parameter.

You:

- **assume** some real number E,
- **solve the differential equation** with that E,
- then **boundary conditions decide** whether that assumed E is allowed.

Most choices of E fail. A discrete set survives. Those survivors are the physical energies.

Why it *looks* like we treat E as "known"

When we write

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}},$$

we are **not** saying E is known.

We are saying:

> "If the energy were E, the wavefunction would behave like this."

Then we **force continuity** of ψ and ψ'.

Those conditions give equations like

$$k\tan(ka) = \kappa.$$

Now substitute back:

$$\sqrt{\frac{2mE}{\hbar^2}}\tan\left(a\sqrt{\frac{2mE}{\hbar^2}}\right) = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}.$$

That is a **nonlinear equation for E**.

So the logic is:

1. Treat E as an unknown constant.
2. Solve the ODE symbolically in terms of E.
3. Boundary conditions turn into an equation **whose unknown is E**.
4. Only discrete solutions exist.

Why energy becomes discrete (the real reason)

If the well were infinite space:

- no boundary conditions,
- any $E > 0$ works,
- continuous spectrum.

With a confining potential:

- boundary conditions overconstrain the solution,
- only certain E allow a nonzero ψ,
- everything else collapses to $\psi = 0$.

This is just **Sturm–Liouville theory in disguise**, wearing a lab coat.

Quick sanity check example

Infinite square well:

$$\psi(x) = A\sin(kx) + B\cos(kx).$$

Boundary conditions:

$$\psi(0) = 0, \quad \psi(L) = 0.$$

These two conditions give:

$$B = 0, \quad \sin(kL) = 0.$$

That second equation is **not solving for ψ**.
It is solving for k, hence for E.

One sentence summary (tattoo this mentally)

You never know E first. You guess E, solve the equation, and the boundary conditions whether that guess was delusional.

And yes, everyone forgets this exact point, including people who later teach it like it was ob

so with all this jargon what is the min E reqd for tunneling place? i am guessing E>V_0?

Short answer: **your guess is wrong**. Which is adorable, in a "classical intuition refusing to c

There is **no minimum energy for tunneling** other than $E > 0$ for a propagating particle to place.

Let's strip the jargon off and pin this down.

Two completely different regimes (don't mix them)

1) Classical "go over the barrier"

- If $E > V_0$: particle goes **over** the barrier.
- Wavevector is real everywhere.
- Transmission $T > 0$, reflection $R > 0$ in general.
- This is **not tunneling**.

2) Quantum tunneling

- If $0 < E < V_0$: particle **classically forbidden** in the barrier.
- Inside barrier:

$$\psi \sim e^{-\kappa x}, \quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

- Exponential decay, but **not zero**.
- Matching conditions force a transmitted wave on the far side.

This is tunneling. Full stop.

So what is the "minimum energy"?

There isn't one.

For **any** E with

$$0 < E < V_0,$$

the transmission probability

$$T = \frac{1}{1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(\kappa L)}$$

is **nonzero**.

As $E \to 0$:

- $\kappa \to \sqrt{2mV_0}/\hbar$,
- $T \to 0$,
- but **never exactly zero**.

Quantum mechanics does not care about your desire for thresholds.

Why $E > V_0$ feels tempting (and wrong)

You're importing classical logic:

> "Particle must have enough energy to cross."

QM replies:

> “Wavefunctions don’t ask permission.”

The barrier doesn’t stop the wave, it **attenuates** it. As long as the wave reaches the far bou
nonzero amplitude, matching conditions *force* transmission.

One clean way to remember it

- $E > V_0$ → over-the-barrier scattering
- $0 < E < V_0$ → tunneling
- **No sharp cutoff** → only exponential suppression

The only thing energy controls is **how embarrassed the transmission probability should l**

If you want a single-line takeaway

Tunneling is not about “enough energy”. It’s about boundary conditions refusing to kill wavefunction completely.

That’s the whole trick.

so how does quantum tunneling happen or take place is always happening for E<V?

Yes. And no. Welcome to the part where language starts lying.

Short version first, then the clean physics.

There is always leakage into a classically forbidden region for $E < V$.
But **tunneling only “happens” when that leakage reaches another allowed region and ha to go.**

Now the real explanation.

What actually happens (not the cartoon)

Take a region where $E < V(x)$.

The Schrödinger equation becomes

$$\psi'' = \kappa^2\psi, \quad \kappa = \sqrt{\frac{2m(V-E)}{\hbar^2}}.$$

That equation **does not allow oscillatory solutions**.
It only allows exponentials:

$$\psi(x) = Ae^{-\kappa x} + Be^{+\kappa x}.$$

So the wavefunction **must decay**, not vanish.

That decay is not optional. It is the *only* way the equation can be satisfied.

So is “leakage” always happening?

Yes, locally

At *any* boundary between:

- a classically allowed region ($E > V$),
- and a forbidden region ($E < V$),

the wavefunction **must penetrate** a distance

$$\ell \sim \frac{1}{\kappa}.$$

This happens even in:

- bound states,
- finite wells,
- atoms,
- harmonic oscillators.

That tail is **not tunneling**. It's just evanescence.

When does it become tunneling?

Tunneling requires **three regions**, not two:

1. Allowed region (incoming wave)
2. Forbidden region (decay)
3. Allowed region again (outgoing wave)

If the forbidden region has **finite width**, the decaying wave reaches the far side with nonze
Continuity then *forces* an outgoing oscillatory wave.

That's tunneling.

If the forbidden region is **semi-infinite**, the wave just dies quietly and nothing comes out. N
just a tail.

Why this is NOT "the particle borrowing energy"

That story should be thrown into a river.

- Energy is conserved.
- The wavefunction never violates the Schrödinger equation.
- There is no classical trajectory inside the barrier.

The particle does **not** go over or through in time.
The stationary state exists everywhere at once.

Tunneling is not a process.
It is a property of the global solution.

One brutally clean way to think about it

- **QM does not ask where the particle is.**
- It solves a differential equation globally.
- Boundary conditions couple regions that classical physics pretends are separate.
- Exponentials do not hit zero at finite distance.
- So amplitude survives.
- So transmission exists.

No drama. No cheating. No metaphysics.

Final mental model (keep this)

- $E < V$ ⇒ exponential decay **always**
- Decay ≠ tunneling
- **Finite barrier + decay + second allowed region = tunneling**
- Probability current stays conserved the whole time

That's it. Once this clicks, tunneling stops feeling mysterious and starts feeling annoyingly

okay so the deal is whether the width is small enough for tunneling to take place and not whether there is some ki threshold for tunneling to happen right>?

Yes. That's the correct conclusion. Lock it in.

There is **no voltage / energy threshold for tunneling**.
The only question is whether the barrier is **thin enough (or low enough)** that the exponent wave doesn't die before it reaches the other side.

Now let me sharpen that so it's unambiguous and future-proof.

What actually controls tunneling

Transmission for $E < V_0$:

$$T \sim e^{-2\kappa L}, \qquad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}.$$

Everything lives inside the exponent.

Three knobs:

1. **Width** L
   Wider barrier → wave dies more → smaller T
2. **Height** $V_0 - E$
   Higher barrier → larger κ → faster decay
3. **Mass** m
   Heavier particle → tunneling gets brutally suppressed

No thresholds. Just exponential humiliation.

Why people talk about "threshold voltage" anyway

That's **engineering slang**, not quantum mechanics.

In real devices:

- currents below some value are unmeasurable,
- noise exists,
- temperature smears distributions,
- contacts matter.

So engineers say "below this voltage, tunneling doesn't happen," which really means:

> "The current is so small we don't care."

Physics says:

> "It's there. You just don't like the number."

One clean sentence to remember

Tunneling is controlled by length scales, not energy thresholds.

Or more brutally:

Quantum mechanics doesn't switch off. It just becomes exponentially rude.

You've got it now.

okay so in that case can you make sense of fn tunneling
knowledge base? that is also a quantum mechanism. eve
the universe is quantum mechanism... then how does fn

Yes, FN tunneling makes perfect sense with exactly the picture you now have. No new mag
uglier geometry, more hype in device physics papers.

Let's translate Fowler–Nordheim into the language you already rebuilt.

What Fowler–Nordheim tunneling actually is

FN tunneling = electrons tunneling through a *tilted* vacuum barrier due to a strong elec

That's it. No extra quantum sauce.

At a metal surface:

- Inside metal: electrons filled up to the **Fermi energy** E_F.
- Outside metal: vacuum level sits ϕ (work function) above E_F.
- Normally: barrier is thick → tunneling negligible.

Apply a strong electric field F:

$$V(x) = \phi - eFx$$

Now the barrier is **triangular**, not rectangular.

Why tunneling suddenly becomes observable

From your earlier conclusion:

> tunneling depends on barrier *width*, not thresholds.

The electric field does two things:

1. **Reduces the effective width** of the barrier
   The classical turning point moves closer to the surface.
2. **Leaves the height mostly unchanged**

The work function ϕ is still there.

So even though $E < \phi$, the barrier becomes *thin enough* that

$$T \sim \exp\left(-2\int \kappa(x)\,dx\right)$$

is no longer microscopic nonsense.

WKB view (the clean way)

In a spatially varying barrier,

$$\kappa(x) = \sqrt{\frac{2m}{\hbar^2}\big(V(x) - E\big)}.$$

Transmission:

$$\boxed{T \approx \exp\left[-2\int_{x_1}^{x_2} \sqrt{\frac{2m}{\hbar^2}(V(x)-E)}\,dx\right]}$$

For FN tunneling:

$$V(x) = \phi - eFx$$

Do the integral (textbook result):

$$T \sim \exp\left(-\frac{4\sqrt{2m}}{3\hbar eF}\,\phi^{3/2}\right)$$

No threshold. Just a savage exponential in $1/F$.

Why FN current looks like it "turns on"

Current density:

$$J \propto F^2 \exp\left(-\frac{B\phi^{3/2}}{F}\right)$$

So as F increases:

- exponent collapses,
- current jumps by orders of magnitude.

Humans see that and scream "threshold".

Physics calmly says: exponential sensitivity.

Why this does NOT contradict earlier statements

- Electrons **do not need** $E > \phi$.
- They **do not climb over** the vacuum barrier.
- The field **does not give them energy to escape**.
- It **reshapes the barrier** so tunneling becomes likely.

Same mechanism as your finite barrier example. Just continuous and slanted.

One sentence that unifies everything

Fowler–Nordheim tunneling is just ordinary tunneling, weaponized by geometry and an

No new principles. Same Schrödinger equation. Same exponential decay. Same refusal of c
mechanics to respect classical intuition.

You're thinking about this correctly now.